



sTMS Interface Functional Specification

Republic Polytechnic – SA 2.0

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	02/05/2022	Initial version of RP SA2.0 IFS	AvePoint
0.2	14/06/2022	Update Interface Specs at Section 3 Interface fields based on the meeting on 9 June 2022, added the following: 3.11 Assessment Component Fields 3.12 Assessment Schedule Fields 3.13 Assessment Grade Fields	AvePoint
0.3	25/06/2022	Update Interface Specs at Section 2 Interface fields and Section 3 Interface fields based on email follow up as below: 2.3 Business Rules 3.6 StudentInfo Fields 3.11 Assessment Component Fields 3.12 Assessment Schedule Fields 3.13 Assessment Grade Fields	AvePoint
0.4	04/07/2022	Revised Section 2.3 Business Rules based on email update from RP dated on 29 June 2022.	AvePoint
0.5	06/07/2022	Revised Section 3 Interface fields based on discussion on 5 July 2022.	AvePoint
0.6	14/07/2022	Revised interface functional specifications based on RP comments on 13 July 2022.	AvePoint
0.7	15/07/2022	Revised interface functional specifications based on RP comments on 14 July 2022.	AvePoint
0.8	18/07/2022	Revised interface functional specifications based on RP comments on 18 July 2022	AvePoint
1.0	19/07/2022	Finalized Version by RP and AvePoint	AvePoint



Document Acceptance Form

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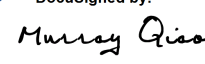
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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS shared Training Management System (sTMS) and Republic Polytechnic SA 2.0 for publishing CET data and student information. sTMS shall interface with RP SA 2.0 to retrieve/send the information listed in the following sections.

1.2 Intended Audience

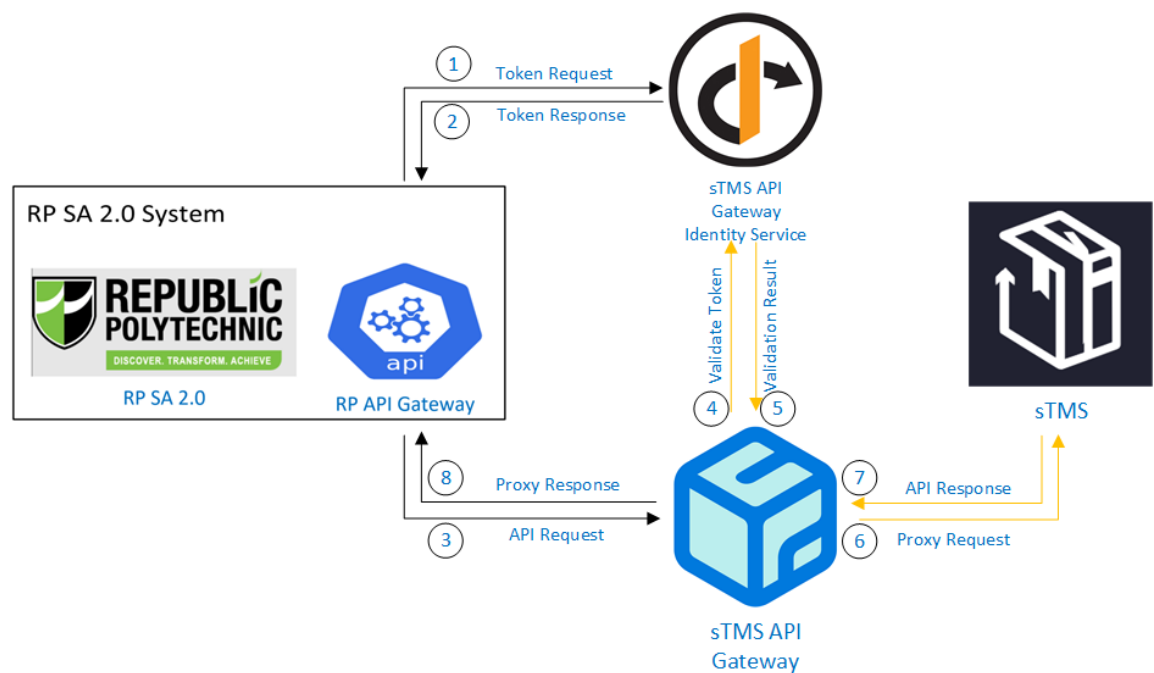
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of both systems.

2. Technical Overview

2.1 System Flow

There are 4 roles involved in the interface diagram.

- **RP SA 2.0 and RP API Gateway:** client application that consumes the API.
- **sTMS API Gateway:** sTMS interface Integration middleware that validates HTTP request authentication, authorization, and payload inspection.
- **sTMS API Gateway Identity Service:** sTMS interface Identity Issuer that issues access token for API authentication and authorization
- **sTMS:** shared Training Management System.



SN	Description
1	RP SA 2.0 API Gateway sends POST request with <i>client_id</i> and <i>secret</i> to sTMS API Gateway Identity Service to get API access token.
2	sTMS API Gateway Identity Service sends the access token back to RP API Gateway within response message.

	This access token can be reused within 1 hour, client application can use an access token for multiple API request before the access token expires, after that client application needs to request a new access token again.
3	RP API Gateway sends HTTP POST request with given student id and reference date to sTMS API Gateway.
4	sTMS API Gateway sends request to sTMS API Gateway Identity Service to validate the access token of the API info request.
5	sTMS API Gateway Identity Service sends back the validation result.
6	sTMS API Gateway verifies the request and sends a proxied request to sTMS.
7	sTMS queries the interface data in database and sends response to API Gateway.
8	sTMS API Gateway sends the response to RP SA 2.0 API Gateway.

2.2 Interface Design Summary

Hosting System	shared Training Management System
Data Source System	shared Training Management System
Data Destination System	Republic Polytechnic SA 2.0 System
Mode of Interface	HTTP API
Method	POST
Content-Type	application/json
Authorization	Bearer <access_token>
Frequency	Realtime
Encryption Method	HTTPS
Endpoint URL	<TBC>

2.3 Business Rules

No.	Business Rule
1	The confirmed invigilator details would be synced from RP SA2.0 to sTMS.



No.	Business Rule
2	Invigilator claims would be triggered from RP SA2.0 to Workday.

2.4 Interface Authentication

sTMS use OAuth (OAuth2.0) to perform authentication. RP SA 2.0 needs to send HTTP POST request to sTMS Identity Service endpoint to get access token and put the access token into HTTP request 'Authorization' to consume target API.

sTMS Identity Service Endpoints:

Env	URL
UAT	<TBC>
Production	<TBC>

To access the API, the client application needs to authenticate using the client credential grant from API gateway for an access token with the following parameters:

- **grant_type**: REQUIRED. The value must be set to "client_credentials"
- **client_id**: REQUIRED. Client ID.
- **client_secret**: REQUIRED. Client Secret.

Sample:

```
curl --location --request POST '{URL}' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'grant_type=client_credentials' \
--data-urlencode 'client_id={client_id}' \
--data-urlencode 'client_secret={client_secret}'
```

Response Body:

```
{
  "access_token": "{access token}",
  "expires_in": 3600,
```




```
"token_type": "Bearer",  
  
"scope": "{api scope}"  
  
}
```

3. Interface Fields

3.1 School Fields

The RP school's creation, updating, or deletion information will be synchronized to SA 2.0 via API call provided by sTMS.

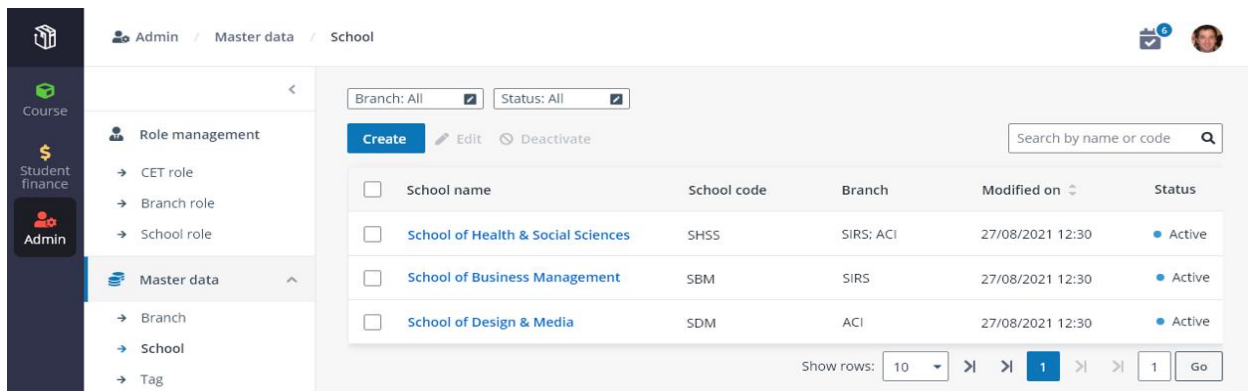
Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
SchoolCode*	ACADEMIC_ORGANISATION	School Code	ACE, SEG	String(\$nvarchar(255))
SchoolName*	DESCRIPTION_FOR_MAL	School Name	School of Health & Social Sciences	String(\$nvarchar(255))
CreateDate*	EFFECTIVE_DATE	Effective Date	2012-04-16	String(\$YYYY-MM-DD)
Status*	EFFECTIVE_STATUS	Effective Status (Active, Inactive)	A, I	String(\$nvarchar(1))
ModifiedOn*	LAST_UPDATE_DATETIME	Last Updated Datetime	2016-03-22T15:39:03	String(\$YYYY-MM-DD hh:mm:ss)
Action*	AUDIT_ACTION	Create/Update/Delete	C, U, D	String(\$nvarchar(1))

The asterisk * represents the field is required.



Admin / Master data / School

Branch: All ☒ Status: All ☒

Create Edit Deactivate

Search by name or code

☐	School name	School code	Branch	Modified on	Status
☐	School of Health & Social Sciences	SHSS	SIRS: ACI	27/08/2021 12:30	Active
☐	School of Business Management	SBM	SIRS	27/08/2021 12:30	Active
☐	School of Design & Media	SDM	ACI	27/08/2021 12:30	Active

Show rows: 10 1 1 Go

3.2 QualificationType Fields

The RP Qualification Type's creation, updating, or deletion information will be synchronized to SA 2.0 via API call provided by sTMS.

Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Filed Name	Description	Sample Value	Data Type
QualificationType*	QUALIFICATION_TYPE	Send the default Qualification Type: CET	CET	String(\$nvarchar(20))
CreateDate*	EFFECTIVE_DATE	Create Date	1901-01-01	String(\$YYYY-MM-DD)
Status*	EFFECTIVE_STATUS	Status (Active, Inactive)	A, I	String(\$nvarchar(1))
Category*	QUALIFICATION_TYPE_CATEGORY	Send default value CET	CET	String(\$nvarchar(20))
ModifiedOn*	LAST_UPDATE_DATETIME	Last Updated Datetime	2016-03-22T15:39:03	String(\$YYYY-MM-DD hh:mm:ss)
Action*	AUDIT_ACTION	Create/Update/Delete	C, U, D	String(\$nvarchar(1))

The asterisk * represents the field is required.

3.3 Semester Fields

The RP semester's creation, updating, or deletion information will be synchronized to SA 2.0 via API call provided by sTMS.

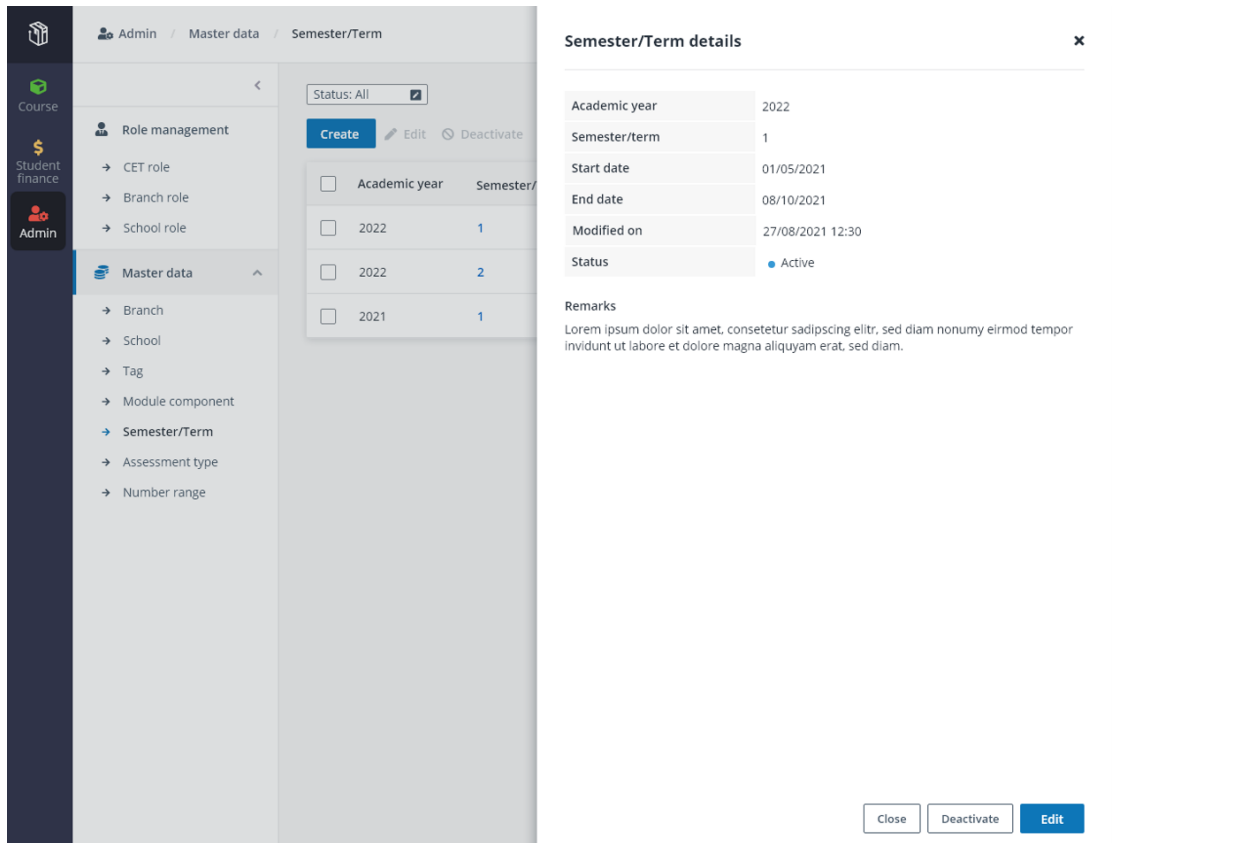
Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
QualificationType*	QUALIFICATION_TYPE	Send Default Value: CET	CET	String(\$nvarchar(20))
SemesterCode*	SEMESTER_CODE	Semester Code	C2201	String(\$nvarchar(20))
StartDate*	SEMESTER_BEGIN_DATE	Semester Start Date	4/17/2012	String(\$YYYY-MM-DD)
EndDate*	SEMESTER_END_DATE	Semester End Date	9/3/2012	String(\$YYYY-MM-DD)
Category*	SEMESTER_CATEGORY	Send Default Value: R	R	String(\$nvarchar(1))
AcademicYear*	ACADEMIC_YEAR	Academic Year	2012, 2013	String(\$YYYY)
SemesterNo*	SEMESTER_NUMBER	Semester Number	1, 2	Number
ModifiedOn*	LAST_UPDATE_DATETIME	Last Updated Datetime	2016-03-22T15:39:03	String(\$YYYY-MM-DD hh:mm:ss)
Action*	AUDIT_ACTION	Create/Update/Delete	C, U, D	String(\$nvarchar(1))

The asterisk * represents the field is required.



The screenshot shows the AvePoint Admin interface. On the left is a sidebar with navigation links: Course, Student finance, and Admin. The Admin section is expanded, showing a list of Master data items: Branch, School, Tag, Module component, Semester/Term (selected), Assessment type, and Number range. The main content area displays the 'Semester/Term details' modal. The modal has a title bar with a close button. Below the title bar, there's a table with the following data:

Academic year	2022
Semester/term	1
Start date	01/05/2021
End date	08/10/2021
Modified on	27/08/2021 12:30
Status	Active

Below the table, there's a 'Remarks' section with a text area containing placeholder text: 'Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam.' At the bottom right of the modal, there are three buttons: 'Close', 'Deactivate', and 'Edit'.

3.4 Programme Fields

The RP programme's creation, updating, or deletion information will be synchronized to SA 2.0 via API call provided by sTMS.

Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
CourseCode*	PROGRAMME	Course Code	DBMEE, IHCM	String(\$nvarchar(255))
CreateDate*	EFFECTIVE_DATE	Create Date	1901-01-01, 2012-04-16	String(\$YYYY-MM-DD)

Status*	EFFECTIVE_STATUS	Effective Status (Active, Inactive)	A, I	String(\$nvarchar(1))
CourseName	DESCRIPTION	Course Name	Dip in Biomed Electronics Eng, MC Int Human Capital Mgt	String(\$nvarchar(max))
QualificationType*	QUALIFICATION_TYPE	Send Default Value: CET	CET	String(\$nvarchar(255))
SchoolCode*	ACADEMIC_ORGANISATION	School Code	SEG, CEC	String(\$nvarchar(255))
ModeOfTraining*	PROGRAMME_TYPE	Mode of Training TPG related courses would be returned Full time and Part time while non-TPG related courses would return as NA.	FT, PT	String(\$nvarchar(20))
CourseName	DESCRIPTION_LONG	Course Name	Diploma in Biomedical Electronics Engineering, Modular Certificate in International Human Capital Management	String(\$nvarchar(255))
CourseManager	PROGRAMME_CHAIR_STAFF_ID	Course Manager	0039, 1145	String(\$nvarchar(255))
CourseCoordinator	ASSISTANT_PROGRAMME_CHAIR_STAFF_ID	Course Coordinator	0040, 1146	String(\$nvarchar(255))
ModifiedOn*	LAST_UPDATE_DATETIME	Last Updated Datetime	2016-03-22T15:39:03	String(\$YYYY-MM-DD hh:mm:ss)
Action*	AUDIT_ACTION	Create/Update/Delete	C, U, D	String(\$nvarchar(1))

The asterisk * represents the field is required.



Course

Student finance

Admin

Course configuration

Course intake configuration

Timetabling configuration

Catalogue configuration

Form Bank

Venue management

Course category: All

Course type:

Create

Import

Export

☐ Course name

☒ MSc e-Business Management

☐ Data Science and Digital Operation

☐ Cloud and Machine learning

☐ Cloud and Machine learning

☐ Cloud and Machine learning

Course

TP-SD-F

TP-SD-F

TP-SD-F

TP-SD-F

TP-SD-F

Assign role

Course manager

Manager who update the course, intake, catalogue information for a particular course

Jonah Louis

Course approver

Approver who review and make application/enrolment approvals for a particular course

Jonah Louis

Course coordinator

Coordinator who make scheduling arrangements for a particular course

Jonah Louis

Course fee manager

Manager who set up and verify fee information on course and intake for a particular course

Jonah Louis

Cancel Save



SSG related

Target audience

Target training group

In-house

Course cluster

Buesiness & Management

Mode of training

Full time and part time

Field of study - SSEC 2015

Course sector

Area of training

SkillsFuture series emerging area

Data analytics

Level of training

PME

SkillsFuture Series Level

Basic

Industry transformation map (ITM)

Built Environment - Cleaning

Medium of instruction

Course

Course configuration

Create course

Version 2.0

Course configuration

General information

Qualification

Course fee

Step 1. General information

Lorem ipsum dolor sit amet, consectetur adipiscing elit. sed diam nuncy enim tempor invidunt ut lobortis et diam magna aliquam; eros, and diam aliquam. At vero eos et

Institution *

(NPT) Nanyang Polytechnic

Branch

(SRS) Singapore Institute of Retail Studies

TIS approved? *

☒ Yes ☐ No

Course name *

MSc e-Business Management

Course code *

Effective from *

12/30/2021

No. of training hours *

50

Course category *

Part time diploma (PTD)

Assessment type

Project

Approval date

06/03/2021

National Manpower Cluster

Business Management and others

Area of interest *

Accounting Banking & Finance

Email

josh.louis@gmail.com

School

(SBM) School of Business Management

TIS course reference No. *

SSG-CIM-123-123-240/023

Course short name

MSBM

Diploma course code

123456

Effective to

12/30/2022

No. of assessment hours

4

Course type *

Academic Full Qualification (FQ)

Senate reference No.

Sunset date

06/03/2023

WSP indicator *

☐ Yes ☒ No ☐ N/A

TIS contact person

Jonah Louis

Phone

+65 1545554

Course objectives

Fusing core concepts of engineering, finance, function, aesthetics and lifestyle, the Diploma in Engineering Design with Business (DEDB) equips you with the know-how to transform your entrepreneurial dreams into a viable business plan.

Through DEDB, you will gain essential skills and knowledge in both business and engineering aspects that will give you a head start to launch your start-up ideas in various industries. You will be exposed to practical learning experience and the opportunity to further hone your creativity when you undergo internship at established companies.

Course description

Thank you for your interest. This course is currently fully subscribed. You can submit your interest by clicking on the "Register Interest" button, and we will contact you when the next course run is available.

SSG related

Target audiences

3 selected

Target training groups

In-house

Course cluster

Business & Management

Field of study - SSEC 2015

Mode of trainings

Full time and part time

3 selected

Course sectors

3 selected

Area of trainings

3 selected

Level of training

PME

SkillFuture series emerging area

2 selected

SkillFuture Series Level

Basic

Industry transformation map (ITM)

3 selected

Other Information

Corporate customisation *

☒ Yes ☐ No

Company name *

Required student accounts

3 selected

Tag

Dolor

Amet

Cancel

Save as draft

Save and end

3.5 Student Enrollment Fields

The creation, updating, or deletion of RP student enrollment information will be synchronized to SA 2.0 via API call provided by sTMS.

Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
StudentId*	STUDENT_ID	Student ID (UAM Common Id)	16000831	String(\$nvarchar(20))
EnrolmentStatus*	ENROLMENT_STATUS	Enrollment Status, Enroll or Drop	E/D	String(\$nvarchar(1))
ModuleCode*	MODULE_ID	Module Code	700178	String(\$nvarchar(20))
SemesterId*	SEMESTER_CODE	Semester Code	C2020	String(\$nvarchar(20))
ClassId*	CLASS_ID	Class ID	GUID	String(\$nvarchar(32))
ClassName*	CLASS_LABEL	Class Name	DN-1-A1009C-A	String(\$nvarchar(255))
EnrollmentId*	Enrollment_ID	Enrollment ID	GUID	String(\$nvarchar(32))

The asterisk * represents the field is required.



Course / Course intake configuration / Timetabling management

Home
Course
Student finance
Admin
Student Record

Course intake setup
Course intake management

Timetabling management
Application management
Enrolment management
Request management

Semester 1
(01/02/2022 - 31/03/2022)

BM0001- Business management
Class 1
Class 2

BM0002- Business fundamentals
Class 1
Class 2

BM0003- Business law
Class 1
Class 2

Semester 2
(01/04/2022 - 31/05/2022)

BM0001- Business

View class session

Session A

Session No.

1

Module name

e-Business Fundamentals

Class name

Class 1

Topic

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Mode of training

classroom

Record attendance

Yes

Class session details

Trainer

Ashley Lim - 1 hours
Ada Liu - 2 hours

Facility

Session start time

01/11/2021 09:00

Session end time

01/11/2021 12:00

Home

Course

Student finance

Admin

Student Record

Course intake setup

Course intake management

Timetabling management

Application management

Enrolment management

Request management

Student enrolment list

Module enrolment

Failed students list

Module exemption / Credit transfer

Deferment

Deferred module return

Withdrawal

Reinstatement

Repeat module

Grade review

LOA

Course / Course intake configuration / Enrolment management / Student enrolment list

TP-SD-FQ0001-220001 - e-Business Management

Course intake No.202101

StatusApplication closed

Intake notificationSend notification

Payment modePay upon application

Course categoryPart-time Diploma (PTD)

Course typeAcademic Full Qualification (FQ)

Course duration01/02/2022 - 31/03/2022

Registration period02/01/2022 - 03/01/2022

Semester 1Semester 2Semester 3

Semester duration: 01/02/2022 - 15/02/2022

Sponsorship: All

Enrolment status: All

TPG course run id: All

Search by student name...

EnrolWithdrawTransferSuspendImportExport

Show all payment status

☐	Name	Student ID	Sponsorship	Payment status	SFC status	PSEA status	GIRO status	Other
☐	AL Ashley Lim	S1234651	Self	Paid	N/A	Disbursed	Paid	Pa
☐	BL Bruce Li	S9452222	Self	Paid	Paid	N/A	Paid	N/
☒	CQ Cindy Qian	S1234578	Company	Unpaid	N/A	N/A	Rejected by bank	Ur
☐	DL Dandy Low	S1233698	Self	Paid	Ready to payout	Disbursed	Paid	Pa
☐	FL Febbie Lin	S1234654	Self	Deemed paid	Approved	Form verified	Form received	Pa
☐	AL Amy Lim	Reallocation required	If	Deemed paid	N/A	Form verified	Form received	Pa
☐	KZ Kevin Zhang	S1234657	Self	Unpaid	Pending	Pending	Rejected by bank	Ur
☐	DW David Wang	S1234659	Self	Unpaid	Rejected	Pending	Rejected by bank	Ur
☐	AL Adam Liu	S1234785	Self	Unpaid	Cancelled	Form verified	Rejected by bank	Pa

3.6 StudentInfo Fields

The RP student’s creation, updating information will be synchronized to SA 2.0 via API call provided by sTMS. After graduation, the student will be updated to inactive in SA2.0 via interface job.

1. Get Student Id List

Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

18

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
StudentIdList	STUDENT_ID	Student ID list		String[]

2. Get Student Details

Http Request Field

Name	Description	Data Type	Remarks
StudentId	The Student Id is used to query student detail information.	String(\$nvarchar(20))	

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
StudentId*	STUDENT_ID	Student ID (UAM Common Id)		String(\$nvarchar(20))
QualificationType*	STU_QUALIFICATION_TYPE	Send default value: CET	CET	String(\$nvarchar(20))
Name*	FULL_NAME	Name		String(\$nvarchar(255))
Gender*	GENDER	Gender: Male, Female	F, M	String(\$nvarchar(1))
Mobile	HANDPHONE	Mobile number		String(\$nvarchar(20))
DateOfBirth	DOB	Date of birth	1/1/1995	String(\$YYYY-MM-DD)
NationalId	NATIONAL_ID	Nationality	China	String(\$nvarchar(30))
Citizenship	Citizenship	Citizenship	Singaporean	String(\$nvarchar(30))
AddressDetails	ADDRESS_DETAILS	Address details, the format is as following: [Block/Building Number] [Street Name] #[Floor No. -Unit No.] [Building Name] [Country code] [Postal Code]		String(\$nvarchar(255))



Email*	EMAIL	Email address		String(\$nvarchar(70))
School*	SCHOOL	School name	ACE, SEG	String(\$nvarchar(10))
StudentStatus*	STUDENT_STATUS	Student Enrolment Status based on RP SA2.0 Active - Enrolled, Transferred Inactive - Deferred, Withdrawn, Completed, Suspended	Active/Inactive	String(\$nvarchar(20))
Programme*	PROGRAMME	Course name	DBMEE, IHCM	String(\$nvarchar(255))
Program_Code	PROGRAM_CODE	Course Code	B06D	String(\$nvarchar(30))
Prog_Status	PROG_STATUS	Student Enrolment Status based on sTMS	Enrolled, Transferred, Deferred, Withdrawn, Completed, Suspended	String(\$nvarchar(255))
CertificateName	Diploma	Certificate Name	DIT	String(\$nvarchar(255))
ETHNIC_GRP_CD	ETHNIC_GRP_CD	NIL	NIL	String(\$nvarchar(8))
AdmissionYear	Admission_Year	Admission Year	2010	Int
MatriculationYear	Matriculation_Year	Matriculation Year	2010	Int
Photo	PHOTO	Student Photo		Byte[]
Action*	AUDIT_ACTION	Create/Update/Delete	C / U	String(\$nvarchar(1))

The asterisk * represents the field is required.

Course

Student finance

Admin

Version 2

<

Course configuration

1 General information

2 Course details

3 Course fee

Course / Course configuration / Create course

4

Step 2. Course details

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et.

Course structure

Course certification

Minimum entry requirement

Course level certification

+ Add certification

Certification	And/Or
Certificate in Business Management	☒ And ☐ Or
Big data analyser certificate	☐ And ☐ Or

Relevant working experience

Other requirements

Certification structure

+ Add certification

Total hours: 240

Course

Specialist diploma in e-Business Management

Post Diploma Certificate

Core

Specialist diploma in e-Business Management

Pathway 1

Module (e-Business Management)

Core

Statement of result

Module (Digital Development with Python)

Elective

None

Post Diploma Certificate

Elective

Specialist diploma in Digital Development with Python

Pathway 2

Module (Digital Development with Python)

Elective

Statement of result

Statement of participation

Cancel

Save as draft

Back

Save and next



Course

Student finance

Admin

Version 2

Course configuration

1 General information

2 Course details

3 Course fee

Step 2. Course details

Course structure

Course certification

Prerequisite

Course level certification

Certification

Relevant working experience

Other requirements

Certification structure

Edit certification

Certificate name *

Specialist diploma in Digital Development with Python

Certificate level *

Post Diploma Certificate

Max candidature period

Pathway to complete *

Pathway 1

☒ Specialisation

Certificate type

☒ Core ☐ Optional ☐ Elective

WSP indicator *

☐ Yes ☒ No ☐ N/A

TPG course reference No.

SSG-CSM-123-123-2XU923

Cancel

Save

Student Record

Students info

Course info

Application

e-Dossier

Home

Course

Student finance

Admin

Student Record

Student Record / David Lim X16103548 / Student info

Name

David Lim

Citizenship type

Singapore citizen

Race

XXXX

Country of birth

Singapore

Date of birth

20/02/1990

NRIC/FIN No.

12*****78

Gender

Male

National service

Completed NS. ORD date

01/01/2030

Mandatory

Yes

Other information

Edit

Contact info

Mobile number

+65 15454554

Email

David.Lim@email.com

Residential address

Singapore

Country code

Singapore

Block/Building number

XXXXX

Street name

Street A

Postal code

XXXXX

Building name

XXXXX

Floor No. - Unit No.

#10-2

Mailing address

Same as residential address

Country code

Singapore

Postal code

XXXXX

3.7 Module Fields

The RP module’s creation, updating, or deletion information will be synchronized to SA 2.0 via API call provided by sTMS.

Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
ModuleId*	MODULE_ID	Module ID would be based on Course intake level	700178	String(\$nvarchar(20))

ModuleCode*	MODULE_CODE	Module Code	M201, S266	String(\$nvarchar(20))
ModuleOffer*	MODULE_OFFER_NUMBER	Module Offer Number, Send default value: 1	1	String(\$nvarchar(1))
CreateDate*	EFFECTIVE_DATE	Effective Date	1/1/1901	String(\$YYYY-MM-DD)
Status*	EFFECTIVE_STATUS	Effective Status (Active, Inactive)	A, I	String(\$nvarchar(1))
ModuleName*	MODULE_TITLE	Module Title	Communication Theories, Performance Analysis	String(\$nvarchar(255))
ModularCredits	MODULE_UNIT_MAXIMUM	Maximum Credit Unit	4	String(\$nvarchar(4))
ModuleSchool	ACADEMIC_ORGANISATION	School Name	CEC, SHL, SEG, STA, SAS	String(\$nvarchar(255))
QualificationType*	QUALIFICATION_TYPE	Send Default Value: CET	CET	String(\$nvarchar(20))
SemesterCode*	SEMESTER_CODE	Semester	C1710, C1810	String(\$nvarchar(5))
SemesterNumber*	SEMESTER_NUMBER	Semester Number	1, 2	String(\$nvarchar(2))
AcademicYear*	ACADEMIC_YEAR	Academic Year	2018	String(\$nvarchar(4))
ModuleManager	MODULE_MANAGER	Staff ID of Module Manager	0041, 1147	String(\$nvarchar(255))
CoModuleManager	CO_MODULE_MANAGER	Staff ID of Module Coordinator	NA	String(\$nvarchar(255))
ModifiedOn*	LAST_UPDATE_DATETIME	Last Updated Datetime	2016-03-22T15:39:03	String(\$YYYY-MM-DD hh:mm:ss)
Action*	AUDIT_ACTION	Create/Update/Delete	C, U, D	String(\$nvarchar(1))

The asterisk * represents the field is required.

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Course / Module configuration / Create module

Nanyang Polytechnic

JL

Version 1.0

Module configuration

Fill up mandatory fields in this step.

1 General information

2 Module details

Dashboard

Course

Student finance

Admin

Student record

Company record

Report

Step 1. General information

Fill up mandatory fields in this step.

Institution *

(NYP) Nanyang Polytechnic

Branch

(SIRS) Singapore Institute of Retail Studies

School

(SBM) School of Business Management

Module name *

Module code *

BM0001

Module component

Lecturer Tutorial

Effective from *

12/30/2021

Effective to

12/30/2022

No. of module hours

120

No. of modular credits

4

Module synopsis

Remarks

Remarks

Owner assignment

Module owner

JL Jonah Louis

Tag

DIST Group

01

Cancel

Save as draft

Save and next



Course

Student finance

Admin

Course / Module configuration

Module configuration

Course configuration

Course intake configuration

Timetabling configuration

Timetabling

Public holiday

Catalogue configuration

Course catalogue

General configuration

Form Bank

Venue management

Module school: All

Effective from

Create

Import

Export

☐ Module name

☒ e-Business Fundamentals

☐ Enterprise e-Commerce Solutions

☐ Digital Development with Python

☐ Big data analytics and Optimisation

Module

BM0001

BM0002

BM0003

BM0004

Assign role

Module manager

Manager who update the module information for a particular module.

JL

Jonah Louis

Course

Student finance

Admin

Version 1.0

Module configuration

1 General information

2 Qualification

Course / Module configuration / Create module

4

Step 1. General information

Fill up mandatory fields in this step. *um dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sea diam voluptua. At vero eos et.*

Institution *

(NYP) Nanyang Polytechnic

Branch

(SIRS) Singapore Institute of Retail Studies

Module name *

Module component

Lecturer x Tutorial x

Effective from *

12/30/2021

No. of module hours

120

Module synopsis

Remarks

Tag

Dolore Amset

School

(SBM) School of Business Management

Module code *

BM0001

Effective to

12/30/2022

No. of modular credits

4

Cancel

Save as draft

Save and next

3.8 Class Fields

The RP class’s creation, updating, or deletion information will be synchronized to SA 2.0 via API call provided by sTMS.

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Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
ModuleCode*	MODULE_ID	Module Code	001009, 666811	String(\$nvarchar(20))
ModuleOffer*	MODULE_OFFER_NUMBER	Module Offer Number Default value = 1	1	String(\$nvarchar(20))
SemesterCode*	SEMESTER_CODE	Semester	1210, 1331	String(\$nvarchar(20))
SessionCode*	SESSION_CODE	Session Code	1	String(\$nvarchar(1))
ClassId*	CLASS_ID	Class ID	GUID	String(\$nvarchar(32))
SessionNo*	CLASS_NUMBER	Class Number	1234, 5678	String(\$nvarchar(4))
Component	CLASS_COMPONENT	Class Component	DG, MSA, ESE	String(\$nvarchar(50))
ClassName*	CLASS_LABEL	Class Label	M201-2-E66H-A	String(\$nvarchar(255))
ModifiedOn*	LAST_UPDATE_DATETIME	Last Updated Datetime	2016-03-22T15:39:03	String(\$YYYY-MM-DD hh:mm:ss)
Action*	AUDIT_ACTION	Create/Update/Delete	C, U, D	String(\$nvarchar(1))

3.9 Session Fields

The RP session's creation, updating, or deletion information will be synchronized to SA 2.0 via API call provided by sTMS.

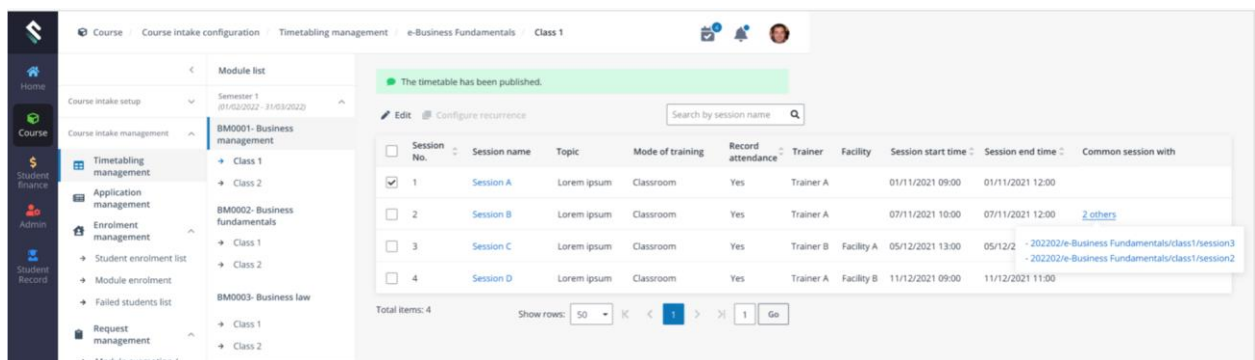
Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
ModuleCode*	MODULE_ID	Module ID	001009, 666811	String(\$nvarchar(20))
ModuleOffer*	MODULE_OFFER_NUMBER	Module Offer Number	1	String(\$nvarchar(1))
SemesterCode*	SEMESTER_CODE	Semester	1210, 1331	String(\$nvarchar(20))
SessionCode*	SESSION_CODE	Session Code	1	String(\$nvarchar(10))
SessionNo*	CLASS_MEETING_NUMBER	Class Meeting Number	1, 2	String(\$nvarchar(1))
SessionId*	CLASS_MEETING_ID	Class Meeting ID	GUID	String(\$nvarchar(32))
Facility*	FACILITY_ID	Facility ID	E66G, W25F	String(\$nvarchar(255))
SessionStartDateTime*	CLASS_MEETING_START_TIME	Session Start Time	18:00:00	String(\$YYYY-MM-DD hh:mm:ss)
SessionEndDateTime*	CLASS_MEETING_END_TIME	Session End Time	19:00:00	String(\$YYYY-MM-DD hh:mm:ss)
Trainer	CLASS_INSTRUCTOR_STAFF_ID	Staff ID of Class Instructor	0866, 00041319	String(\$nvarchar(255))
ModifiedOn*	LAST_UPDATE_DATETIME	Last Updated Datetime	2016-03-22T15:39:06	String(\$YYYY-MM-DD hh:mm:ss)
Action*	AUDIT_ACTION	Create/Update/Delete	C, U, D	String(\$nvarchar(1))

The asterisk * represents the field is required.



The screenshot displays the 'Timetabling management' interface in sTMS. A green notification banner at the top states 'The timetable has been published.' Below this, a table lists sessions with columns: Session No., Session name, Topic, Mode of training, Record attendance, Trainer, Facility, Session start time, Session end time, and Common session with. The table contains four rows of session data. A search bar and pagination controls are visible at the bottom of the table.

Session No.	Session name	Topic	Mode of training	Record attendance	Trainer	Facility	Session start time	Session end time	Common session with
1	Session A	Lorem ipsum	Classroom	Yes	Trainer A		01/11/2021 09:00	01/11/2021 12:00	
2	Session B	Lorem ipsum	Classroom	Yes	Trainer A		07/11/2021 10:00	07/11/2021 12:00	2 others
3	Session C	Lorem ipsum	Classroom	Yes	Trainer B	Facility A	05/12/2021 13:00	05/12/2021 15:00	202202/e-Business Fundamentals/class1/session3
4	Session D	Lorem ipsum	Classroom	Yes	Trainer A	Facility B	11/12/2021 09:00	11/12/2021 11:00	202202/e-Business Fundamentals/class1/session2

3.10 Leave Of Absence Fields

The RP LOA's creation, updating, or deletion information will be synchronized to SA 2.0 via API call provided by sTMS.



Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
StudentID*	STU_ID	Student ID		String(\$nvarchar(20))
RequestId*	LOA_APPLICATION_ID	LOA Request ID		String(\$nvarchar(255))
StartDate*	LOA_STARTDATE	LOA Start Date		String(\$YYYY-MM-DD hh:mm:ss)
EndDate*	LOA_ENDDATE	LOA End Date		String(\$YYYY-MM-DD hh:mm:ss)
Comment	COMMENT	Comments		String(\$nvarchar(max))
Result*	STATUS	Status: A – Approved R – Rejected D – Draft	A, R	String(\$nvarchar(1))

The asterisk * represents the field is required.

 All courses
 Dashb...
 My course
 Applic...
 Requests & Task
 Transa...
 Certific...
 Calendar
 Resour...

Requests & Task / Raise request / Raise LOA for sessions request / View details




← View details

Instructions
View history

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et.

Request ID	LOAS_20220202101832	Requested on	25/12/2019 02:35
Student name	Alice Wang	Student ID	T834561
Email	Alice@gmail.com	Request status	Completed

NYP-SD-FQ0001 - e-Business Management

Course intake No.	NYP-SD-FQ0001-220001	Course category	Part-time Diploma (PTD)
Enrolment status	Enrolled	Course type	Academic Full Qualification (FQ)

Requested Sessions

time	Session end time	Course	Module	Result	Comment
:00	03/02/2022 12:00	NYP-SD-SC0001-e-Business Management	e-Business	Approved	Met criteria
:00	03/02/2022 12:00	NYP-SD-SC0001-e-Business Management	e-Business	Rejected	Not met

3.11 Assessment Component Fields

The RP Assessment Component's creation, updating, or deletion information will be synchronized to SA 2.0 via API call provided by sTMS. In sTMS, component and subcomponent would be created in Assessment setting under module level. The RP SA2.0 will autogenerate Store name and Store Setting Name in RP SA Store upon receiving information from sTMS.

Http Request Field

Name	Description	Data Type	Remarks
ReferenceDate	The reference date is used to filter the data and only sent if the modification time is later than the reference date.	String(\$YYYY-MM-DD hh:mm:ss)	If ReferenceDate is empty, the API will return full data.

Http Response Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
Id*	ID	This is the primary key field.	GUID	String(\$nvarchar(32))

The asterisk * represents the field is required.

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3.12 Assessment Schedule Fields

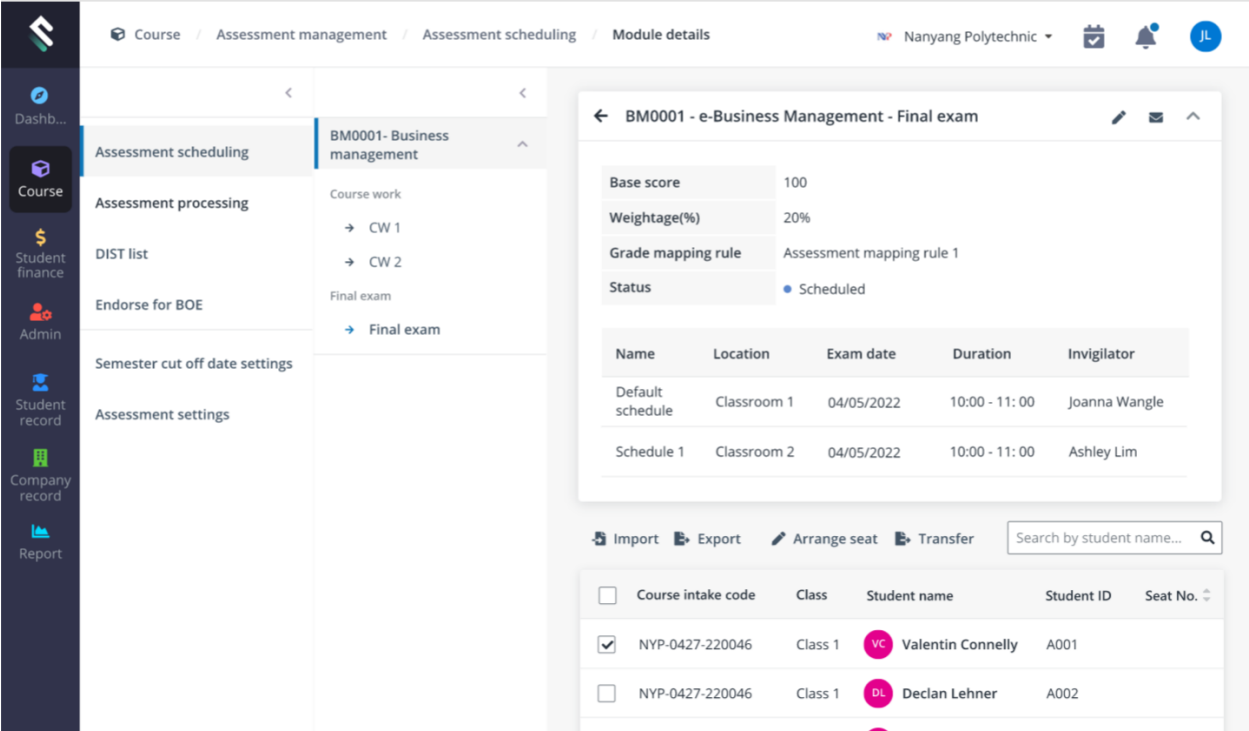
The RP Assessment Schedule's creation, updating, or deletion information will be synchronized from SA 2.0 via API call provided by sTMS. The Assessment Schedule will be first synced from RP SA2.0 to sTMS with the available information from Scheduling Process in Stage 2. With the available and ready invigilator details from Stage 3 Invigilator Assignment, the complete assessment schedule would be synced from RP SA2.0 to sTMS. This exam schedule shall be send in as 'Published' status in sTMS.

The Grade Assignment (GA) from SA2.0 sync to sTMS will not include invigilator, seat number and start time. The Written Assessment will sync all the data listed down in the table.

Http Request Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
Id*	ID	This is the primary key field.	GUID	String(\$nvarchar(32))
StudentID*	STUDENT_ID	This is the primary key field.	15032516	String(\$nvarchar(20))
Semester*	TERM	This is the primary key field.	2022-01	String(\$nvarchar(20))
ModuleCode*	MODULE_CODE	This is the primary key field.	T225	String(\$nvarchar(20))
AssessmentSubComponent*	ASSESSMENT	This is the primary key field.	CW1, EXM1	String(\$nvarchar(20))
StartTime	ASSESSMENT_START_DATE_TIME	Assessment Start Date Time. Format YYYY-MM-DDThh24:mm:ss	2022-08-29T09:00:00	String(\$DateTime)
EndTime	ASSESSMENT_END_DATE_TIME	Assessment End Date Time. Format YYYY-MM-DDThh24:mm:ss	2022-08-29T09:00:00	String(\$DateTime)
ExamLocation	ASSESSMENT_VENUE	Assessment Venue	E24A	String(\$nvarchar(20))
SeatNo	SEAT_NO	Seat number	22	Int
Invigilator	Invigilator	Invigilator of the assessment		String(\$nvarchar(255))
ModifyTime	LAST_UPDATED_DATE_TIME	Format YYYY-MM-DDThh24:mm:ss	2022-01-03T14:10:11	String(\$DateTime)
Remediation	Remediation	Remediation assessment indication	Y, N	String(\$nvarchar(1))
Action	ACTION	C, U, D	C	String(\$nvarchar(255))

The asterisk * represents the field is required.



3.13 Assessment Grade Fields

The RP Assessment Grade's creation, updating, or deletion information will be synchronized from SA 2.0 via API call provided by sTMS. The published grades and scores in RP SA2.0 would be synced to sTMS. This grades and scores shall be marked as 'Confirmed' status in sTMS.

Http Request Field

sTMS Field Name	SA2.0 Field Name	Description	Sample Value	Data Type
Id*	ID	This is the primary key field.	Guid	String(\$nvarchar(32))
Semester*	TERM	Semester Code	2022-01	String(\$nvarchar(20))
ModuleCode*	MODULE_CODE	Module Code	M304	String(\$nvarchar(20))
ClassName	CLASS_CODE	Class Code	Class1	String(\$nvarchar(20))
StudentId*	STU_ID	Student ID	U1234567	String(\$nvarchar(20))
MarkScore*	STU_MODULE_MARK	Mark value	25	Int
MarkGrade	STU_MODULE_GRADE	Grade value	B	String(\$nvarchar(1))
AssessmentSubComponent*	STU_MODULE_ASSESSMENT_TYPE	Assessment Type	CW1	String(\$nvarchar(20))



Remediation	Remediation	Remediation assessment indication	Y, N	String(\$nvarchar(1))
Action*	AUDIT_ACTION	Create/Update/Delete	C, U, D	String(\$nvarchar(1))

The asterisk * represents the field is required.

Dashb...

Course

Student finance

Admin

Student record

Company record

Report

Assessment scheduling

Assessment processing

Cut off date settings

Assessment settings

Endorse for BOE

BM0001- Business management

Module

→ Module overview

Course work

→ Component overview

→ CW 1

→ CW 2

Final exam

→ Final exam

← BM0001 - e-Business Management - CW1

↗

Base score

100

Weightage(%)

20%

Grade mapping rule

Grade mapping rule 1

Unsubmitted

Pending approval

Confirmed

All

Assessment status: All

Search by student name...

✎ Edit

📄 Import

📄 Export

Student ID	Class	Original score	Given score	Score percentage	Mapped grade
A001	Class 1				
A002	Class 1	90.9	90.9	90.9%	A
A003	Class 2	69.1	69.1	69.1%	D
A004	Class 2				

Total items: 4

Show rows: 10

⏪

<

1

>

⏩

1

Go

4. Error Handling

4.1 Error Handling

The table below provides the detailed error listing and handling procedure.

Error Code	Description	Handling Procedure
403	The request is forbidden to access the specified resource.	Verify authorization header in your HTTP request; Raise a ticket to sTMS support to verify the application registration in API gateway.
500	Internal server error	Raise a ticket to sTMS support for further investigation.



5. Assumption and Constraints

5.1 Assumption

No.	Assumptions
1	Datetime values default to Singapore Standard Time (SST). (UTC+08:00)

6. Interface Schemas

Interface Schemas of each API are detailly described in the following table:

School ▼ { SchoolName* string CreateDate* string(\$date-time) Status* string ModifiedOn* string(\$date-time) Action* string }
QualificationType ▼ { CertificateName* string CreateDate* string(\$date-time) Status* string Category* string ModifiedOn* string(\$date-time) }

Semester ▾ {

CertificateName*	string
SemesterCode*	string
StartDate*	string(\$date-time)
EndDate*	string(\$date-time)
Category*	string
AcademicYear*	string
SemesterNo*	string
ModifiedOn*	string(\$date-time)
Action*	string

}

```
Programme ▼ {
  CourseName*      string
  CreateDate*      string($date-time)
  Status*          string
  CourseDescription string
                  nullable: true

  Certificate*     string
  School*          string
  CourseType*      string
  CourseObjective  string
                  nullable: true

  CourseManager    string
                  nullable: true

  CourseCoordinator string
                  nullable: true

  ModifiedOn*      string
  Action*          string
}
```

```
StudentEnrollment ▼ {  
    EnrollmentId*      string  
    StudentId*         string  
    EntrolmentStatus*  string  
    ModuleCode*        string  
    SemesterId*        string  
    ClassSection*      string  
    ClassName*         string  
  
}
```

```
StudentInfo ▼ {  
    StudentId*      string  
    Certificate*     string  
    Name*           string  
    Gender*         string  
    Mobile          string  
                   nullable: true  
    Email*          string  
    School*         string  
    Status*         string  
    Photo           string($byte)  
                   nullable: true  
    Action*         string  
}
```

Module ▼ {

ModuleId*	string
ModuleCode*	string(\$date-time)
CreateDate*	string(\$date-time)
Status*	string(\$date-time)
ModuleName*	string
ModularCredits	string
	<i>nullable: true</i>
ModuleSchool	string
	<i>nullable: true</i>
Certification*	string
SemesterCode*	string
SemesterNumber*	string
AcademicYear*	string
ModuleManager	string
	<i>nullable: true</i>
ModifiedOn*	string(\$date-time)
Action*	string

}

```
Class ▼ {  
    ClassId*           string  
    ModuleCode*        string  
    ModuleOffer*       string  
    SemesterCode*      string  
    SessionCode*       string  
    ClassSection*      string  
    SessionNo*         string  
    Component          string  
                      nullable: true  
    ClassCode*         string  
    ModifiedOn*        string($date-time)  
    Action*           string  
  
}
```

Session ▼ {

ModuleCode*	string
ModuleOffer*	string
SemesterCode*	string
SessionCode*	string
SessionId*	string
ClassSection*	string
SessionNo*	string
Facility*	string
SessionStartDateTime*	string(\$date-time)
SessionEndDateTime*	string(\$date-time)
Trainer	string
	<i>nullable: true</i>
ModifiedOn*	string(\$date-time)
Action*	string

}

LeaveOfAbsense ▼ {

```
StudentID*      string
RequestId*      string
StartDate*      string($date-time)
EndDate*        string($date-time)
Comment         string
                nullable: true

Result*         string
```

```
}
```

AssessmentComponent ▼ {

```
Id*             string
ModuleCode      string
AssessmentComponent* string
AssessmentSubComponent* string
Action*         string
```

```
}
```



```
AssessmentSchedule {  
  StudentList* > [...]  
  Id* string  
  ModuleCode* string  
  AssessmentSubComponent* string  
  StartTime* string($date-time)  
  EndTime* string($date-time)  
  ExamLocation string  
  nullable: true  
  Invigilator string  
  nullable: true  
  ModifyTime string($date-time)  
  Action* string  
}
```

```
AssessmentGrade {
  Id* string
  Semester* string
  ModuleCode* string
  ClassName* string
  StudentId* string
  MarkScore* string
  MarkGrade* string
  AssessmentSubComponent* string
  Action* string
}
```

7. Interface Entities Mapping List

The mapping relationship between sTMS and RP SA 2.0 entities are described in the following table:

sTMS Entities	RP SA 2.0 Entities
School	School
Semester	Semester
Course	Programme
StudentInfo	StudentInfo
Module	ModuleBase
	ModuleInstance
Class	Class
StudentEnrolment	StudentEnrolment
Session	ClassMeeting
	ClassMeetingInstructor
LeaveOfAbsence	LeaveOfAbsence
AssessmentSetting	SaStore
AssessmentProcessing	AssessmentGrade
AssessmentSchedule	AssessmentSchedule